

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

***CO3:** Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 20%

QUESTIONS: 5

**Time Duration: 1 Hour
Total Marks:30**

Annexure A

Resolution Algorithm Implementation

Code of Conduct:

*I Nalluri Akhilesh Kumar(192365074) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

N Akhilesh Kumar

Signature of the student:

Question 1 – Rain and Wet Ground

Knowledge Base

1. If it rains, then the ground becomes wet.
2. It is raining.

Goal

Prove that **the ground is wet** using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF.

Negate the goal.

1. Apply the Resolution Algorithm step by step.
2. Show the Empty Clause ($\square$) if the goal is proved.

The knowledge base contains two statements: if it rains, the ground becomes wet, and it is currently raining. The objective is to prove that the ground is wet using the **Resolution Algorithm**. Resolution works by converting the knowledge base into CNF, negating the goal, and deriving a contradiction in the form of the **Empty Clause** ($\square$).

1. Propositional Logic Representation

Let:

- **R** = It is raining.
- **W** = The ground is wet.

The first statement is:

If it rains, then the ground becomes wet.

In propositional logic:

$$\mathbf{R \rightarrow W}$$

The second statement is:

$$\mathbf{R}$$

The goal is:

$$\mathbf{W}$$

Therefore, the knowledge base is:

$$\mathbf{KB = \{R \rightarrow W, R\}}$$

2. Conversion into CNF

The implication rule is:

$$\mathbf{P \rightarrow Q \equiv \neg P \vee Q}$$

Therefore,

$$\mathbf{R} \rightarrow \mathbf{W}$$

becomes:

$$\neg \mathbf{R} \vee \mathbf{W}$$

So the CNF clauses are:

$$\mathbf{C}_1 = \neg \mathbf{R} \vee \mathbf{W}$$

$$\mathbf{C}_2 = \mathbf{R}$$

3. Negation of the Goal

The goal is:

$$\mathbf{W}$$

For resolution refutation, we negate the goal:

$$\neg \mathbf{W}$$

Therefore, the complete set of clauses is:

$$\mathbf{C}_1 = \neg \mathbf{R} \vee \mathbf{W}$$

$$\mathbf{C}_2 = \mathbf{R}$$

$$\mathbf{C}_3 = \neg \mathbf{W}$$

4. Resolution Algorithm

First, resolve $\mathbf{C}_1$ and $\mathbf{C}_2$.

$$\mathbf{C}_1: \neg \mathbf{R} \vee \mathbf{W}$$

$$\mathbf{C}_2: \mathbf{R}$$

The complementary literals are $\neg \mathbf{R}$ and $\mathbf{R}$. Cancel them:

$$(\neg \mathbf{R} \vee \mathbf{W}) + (\mathbf{R}) \Rightarrow \mathbf{W}$$

Therefore, we derive:

$$\mathbf{C}_4 = \mathbf{W}$$

Now resolve the derived clause $\mathbf{C}_4 = \mathbf{W}$ with the negated goal:

$$\mathbf{C}_4 = \mathbf{W}$$

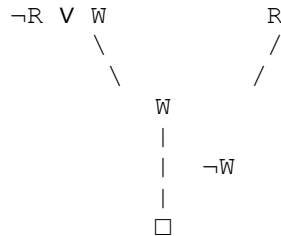
$$\mathbf{C}_3 = \neg \mathbf{W}$$

The complementary literals are **W** and $\neg W$.

Therefore:

$$W + \neg W \Rightarrow \square$$

Resolution Tree



Conclusion

The resolution process produces the **Empty Clause** ($\square$). Therefore, the negation of the goal leads to a contradiction, which proves the original goal.

Hence, the ground is wet.

Question 2 – Student Assignment Submission

Knowledge Base

1. If a student submits the assignment, then the student receives internal marks
2. Rahul submitted the assignment.

Goal

Prove that **Rahul receives internal marks** using the Resolution Algorithm.

Tasks

1. Write the propositional logic expressions.
2. Convert the premises into CNF.

Negate the goal.

1. Apply the Resolution Algorithm.
2. Conclude whether the goal is proved.

The knowledge base states that if a student submits an assignment, the student receives internal marks. It is also given that Rahul submitted the assignment. The objective is to prove that Rahul receives internal marks using resolution.

1. Propositional Logic Representation

Let:

S = Rahul submitted the assignment.

M = Rahul receives internal marks.

The first statement is:

If Rahul submits the assignment, then Rahul receives internal marks.

Therefore:

$$\mathbf{S \rightarrow M}$$

The second statement is:

$$\mathbf{S}$$

The goal is:

$$\mathbf{M}$$

Thus:

$$\mathbf{KB = \{S \rightarrow M, S\}}$$

2. Conversion into CNF

Using:

$$\mathbf{P \rightarrow Q \equiv \neg P \vee Q}$$

we get:

$$\mathbf{S \rightarrow M \equiv \neg S \vee M}$$

Therefore:

$$\mathbf{C_1 = \neg S \vee M}$$

The fact:

$$\mathbf{C_2 = S}$$

is already a CNF clause.

3. Negate the Goal

The goal is:

$$\mathbf{M}$$

Negating the goal gives:

$$C_3 = \neg M$$

Therefore, the clauses used for resolution are:

$$C_1 = \neg S \vee M$$

$$C_2 = S$$

$$C_3 = \neg M$$

4. Resolution Process

Resolve C_1 and C_2 :

$$(\neg S \vee M) + S$$

The complementary literals are $\neg S$ and S .

Therefore:

$$M$$

is derived.

So:

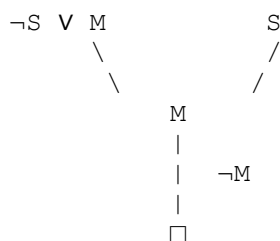
$$C_4 = M$$

Now resolve $C_4 = M$ with the negated goal $C_3 = \neg M$:

$$M + \neg M \Rightarrow \square$$

The Empty Clause is obtained.

Resolution Tree



Conclusion

Since the resolution process produces the Empty Clause $\square$, the negation of the goal is contradictory. Therefore, the original goal is proved.

Hence, Rahul receives internal marks.

Question 3 – Library Membership

Knowledge Base

1. If a person is a library member, then the person can borrow books.
2. Priya is a library member.

Goal

1. Prove that **Priya can borrow books** using the Resolution Algorithm.

Tasks

1. Represent the knowledge base using propositional logic.
2. Convert each statement into CNF.

Negate the goal.

1. Perform resolution until no new clauses can be generated.
2. State the final conclusion.

The knowledge base states that if a person is a library member, the person can borrow books. It is also given that Priya is a library member. The objective is to prove that Priya can borrow books using the Resolution Algorithm.

1. Propositional Logic Representation

Let:

L = Priya is a library member.

B = Priya can borrow books.

The first statement becomes:

$L \rightarrow B$

The given fact is:

L

The goal is:

B

Therefore:

$KB = \{L \rightarrow B, L\}$

2. CNF Conversion

Using the implication equivalence:

$P \rightarrow Q \equiv \neg P \vee Q$

we convert:

$L \rightarrow B$

into:

$$\neg L \vee B$$

Therefore, the CNF clauses are:

$$C_1 = \neg L \vee B$$

$$C_2 = L$$

3. Negate the Goal

The goal is:

$$B$$

Therefore, the negated goal is:

$$C_3 = \neg B$$

The complete resolution set becomes:

$$C_1 = \neg L \vee B$$

$$C_2 = L$$

$$C_3 = \neg B$$

4. Resolution Algorithm

Resolve C_1 and C_2 :

$$(\neg L \vee B) + L$$

The complementary literals are $\neg L$ and L .

Therefore:

$$B$$

is derived.

Thus:

$$C_4 = B$$

Now resolve $C_4 = B$ with the negated goal:

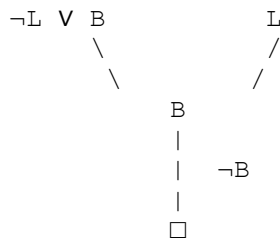
$$C_3 = \neg B$$

Therefore:

$$B + \neg B \Rightarrow \square$$

The Empty Clause is obtained.

Resolution Tree



Final Conclusion

The resolution process generates the Empty Clause $\square$. Hence the negated goal $\neg B$ is inconsistent with the knowledge base.

Therefore, Priya can borrow books.

The result is logically valid because the given fact **Priya is a library member** satisfies the condition required by the rule.

Question 4 – Placement Eligibility

Knowledge Base

1. If a student clears the aptitude test, then the student is eligible for placement.
2. Arun cleared the aptitude test.

Goal

1. Prove that **Arun is eligible for placement** using the Resolution Algorithm.

Tasks

1. Convert the statements into propositional logic.
2. Write the CNF clauses.
3. Negate the goal.
4. Apply the Resolution Algorithm step by step.
5. Draw the resolution tree and conclude.

The knowledge base states that if a student clears the aptitude test, the student becomes eligible for placement. It is given that Arun cleared the aptitude test. The objective is to prove that Arun is eligible for placement using the Resolution Algorithm.

1. Propositional Logic Representation

Let:

A = Arun cleared the aptitude test.

E = Arun is eligible for placement.

The first statement becomes:

$$\mathbf{A \rightarrow E}$$

The given fact is:

$$\mathbf{A}$$

The goal is:

$$\mathbf{E}$$

Therefore:

$$\mathbf{KB = \{A \rightarrow E, A\}}$$

2. CNF Clauses

Using:

$$\mathbf{P \rightarrow Q \equiv \neg P \vee Q}$$

we convert:

$$\mathbf{A \rightarrow E}$$

into:

$$\mathbf{\neg A \vee E}$$

Therefore:

$$\mathbf{C_1 = \neg A \vee E}$$

The fact:

$$\mathbf{C_2 = A}$$

is already in CNF.

3. Negate the Goal

The goal is:

E

Therefore, its negation is:

$$\mathbf{C_3 = \neg E}$$

The complete set of clauses is:

$$\mathbf{C_1 = \neg A \vee E}$$

$$\mathbf{C_2 = A}$$

$$\mathbf{C_3 = \neg E}$$

4. Resolution Algorithm – Step by Step

First, resolve:

$$\mathbf{C_1 = \neg A \vee E}$$

with:

$$\mathbf{C_2 = A}$$

The complementary literals are $\neg A$ and A .

Therefore:

$$\mathbf{(\neg A \vee E) + A \Rightarrow E}$$

So we derive:

$$\mathbf{C_4 = E}$$

Now resolve:

$$\mathbf{C_4 = E}$$

with the negated goal:

$$\mathbf{C_3 = \neg E}$$

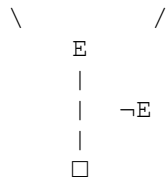
Therefore:

$$\mathbf{E + \neg E \Rightarrow \square}$$

The Empty Clause is obtained.

Resolution Tree

$$\begin{array}{cc} \neg A \vee E & A \\ & \backslash \quad / \end{array}$$



Interpretation of the Result

The Empty Clause indicates that assuming **Arun is not eligible for placement** results in a contradiction with the given knowledge base. Therefore, the original goal must be true.

Conclusion

Hence, Arun is eligible for placement.

The Resolution Algorithm successfully proves the goal because Arun's aptitude-test result satisfies the condition of the placement eligibility rule.

Question 5 – Access Control System

Knowledge Base

1. If a user enters the correct password, then the user is authenticated.
2. If a user is authenticated, then the user is granted access.
3. The user entered the correct password.

Goal

Prove that **the user is granted access** using the Resolution Algorithm.

Tasks

1. Represent the knowledge base in propositional logic.
2. Convert all implications into CNF.
3. Negate the goal.
4. Apply the Resolution Algorithm to derive the Empty Clause ($\square$).
5. Explain each resolution step and write the final conclusion.

The access control system contains two dependent rules. First, entering the correct password makes the user authenticated. Second, an authenticated user is granted access. The user has entered the correct password. Therefore, the objective is to prove that the user is granted access using resolution.

This problem requires **two resolution stages**, because the final conclusion depends on an intermediate fact: **Authenticated**.

1. Propositional Logic Representation

Let:

P = The user entered the correct password.

A = The user is authenticated.

G = The user is granted access.

The first rule is:

If a user enters the correct password, then the user is authenticated.

Therefore:

$$\mathbf{P \rightarrow A}$$

The second rule is:

If a user is authenticated, then the user is granted access.

Therefore:

$$\mathbf{A \rightarrow G}$$

The given fact is:

$$\mathbf{P}$$

The goal is:

$$\mathbf{G}$$

Therefore:

$$\mathbf{KB = \{P \rightarrow A, A \rightarrow G, P\}}$$

2. Convert All Implications into CNF

Rule 1

Original:

$$\mathbf{P \rightarrow A}$$

Using:

$$\mathbf{X \rightarrow Y \equiv \neg X \vee Y}$$

we get:

$$\mathbf{\neg P \vee A}$$

Therefore:

$$C_1 = \neg P \vee A$$

Rule 2

Original:

$$A \rightarrow G$$

Using the same implication rule:

$$\neg A \vee G$$

Therefore:

$$C_2 = \neg A \vee G$$

Fact

The user entered the correct password:

$$P$$

Therefore:

$$C_3 = P$$

All clauses are now in CNF:

$$C_1 = \neg P \vee A$$

$$C_2 = \neg A \vee G$$

$$C_3 = P$$

3. Negate the Goal

The goal is:

$$G$$

For resolution refutation, negate the goal:

$$\neg G$$

Therefore:

$$C_4 = \neg G$$

The complete set of clauses is:

$$C_1 = \neg P \vee A$$

$$C_2 = \neg A \vee G$$

$$C_3 = P$$

$$C_4 = \neg G$$

4. Apply the Resolution Algorithm

Resolution Step 1 – Derive Authentication

Take:

$$C_1 = \neg P \vee A$$

and:

$$C_3 = P$$

The complementary literals are:

$$\neg P \text{ and } P$$

Resolve them:

$$(\neg P \vee A) + P \Rightarrow A$$

Therefore:

$$C_5 = A$$

This means:

The user is authenticated.

Resolution Step 2 – Derive Access

Now use the newly derived fact:

$$C_5 = A$$

with:

$$C_2 = \neg A \vee G$$

The complementary literals are:

A and $\neg A$.

Therefore:

$$A + (\neg A \vee G) \Rightarrow G$$

So:

$$C_6 = G$$

This means:

The user is granted access.

Resolution Step 3 – Derive the Empty Clause

We have derived:

$$C_6 = G$$

But we initially added the negated goal:

$$C_4 = \neg G$$

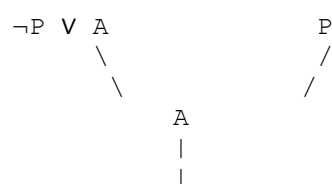
Resolve them:

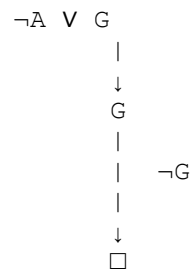
$$G + \neg G \Rightarrow \square$$

Therefore, the **Empty Clause** ($\square$) is obtained.

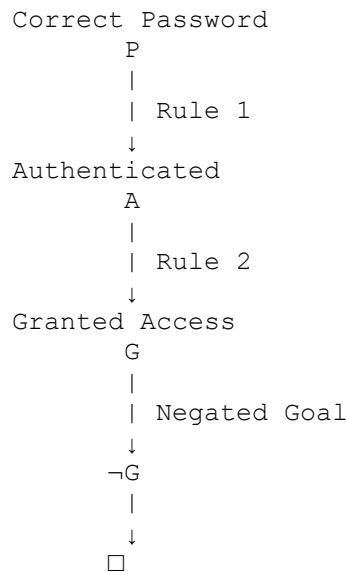
Resolution Tree

The complete reasoning can be represented as:





Or as a complete logical chain:



5. Explanation of Each Resolution Step

The first resolution step uses the fact that the user entered the correct password together with the first rule. This derives **Authenticated**.

The second resolution step uses the derived fact **Authenticated** together with the second rule. This derives **Granted Access**.

Finally, because the goal was **Granted Access**, we introduced its negation $\neg$ **Granted Access**. Since the knowledge base derives **Granted Access**, the two complementary clauses **G** and $\neg$ **G** resolve to the Empty Clause $\square$.

The Empty Clause proves that the negated goal is impossible under the given knowledge base.

Final Conclusion

Therefore:

The user entered the correct password $\rightarrow$ The user is authenticated $\rightarrow$ The user is granted access.

Hence, using the **Resolution Algorithm**, we successfully prove that:

The user is granted access.